

Business Problem Statements

A retail company wants a better handle on how their customers shop, with the goal of improving sales, customer satisfaction, and loyalty over time. Management has noticed shifts in purchasing patterns across different demographics, product categories, and sales channels (online vs. offline), and they want to understand what's actually driving those shifts (things like discounts, reviews, seasonality, or payment preferences) and how those factors relate to repeat purchases.

The task is to dig into the company's consumer behavior dataset to answer one core question:

-How can the company use its shopping data to spot trends, improve customer engagement, and sharpen its marketing and product strategy?

Deliverables:

1. **Data Preparation (using Python)** : Clean and transform the raw dataset so it's ready for analysis.
2. **Data Analysis (using SQL)** : Structure the data, simulate business transactions, and write queries to pull out insights on customer segments, loyalty, and what drives purchases.
3. **Visualization (using Power BI)** : Build an interactive dashboard that surfaces the key patterns and trends for stakeholders.
4. **Report & Presentation** : Summarize the findings and recommendations in a report, and put together a presentation that communicates the insights clearly.